Sayan Saha

Curriculum Vitae

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Education

Aug 2019 – Sep 2024 **PhD in Physics**, *Indian Institute of Science Education and Research (IISER), Pune, India*
Thesis: *Novel Bayesian Inferences from the Cosmic Microwave Background*
Advisor: Prof. Tarun Souradeep.

2017 – 2019 **M.Sc. in Physics**, *IISER, Pune, India*

2014 – 2017 **B.Sc. in Physics**, *Ramakrishna Mission Residential College, Narendrapur (University of Calcutta), India*

Experience

Oct 2024 – Present **Postdoctoral Research Associate**, *Northeastern University, Boston, MA*
Postdoc Advisor: Prof. Jacqueline McCleary.

Sep 2023 – Sep 2024 **Research Fellow (Exchange during PhD)**, *Raman Research Institute, Bengaluru, India*
Host: Prof. Tarun Souradeep.

Sep 2022 – Aug 2023 **Swiss Government Excellence Fellow (Exchange during PhD)**, *Université de Genève, Switzerland*
Host: Prof. Julien Carron.

Publications

First- and Second-Author Publications

2026 **S. Saha**, M. Amit, J. McCleary et al., *Lensing in the Blue IV: The First Weak Gravitational Lensing Maps from the Stratosphere*, arXiv:2610.07244 [astro-ph.CO], submitted to ApJ Letters.

2026 **S. Saha**, J. McCleary et al., *Lensing in the Blue III: Weak Lensing Shape Catalogs of 30 Merging Galaxy Clusters*, *Astrophys. J.* **1011** (2026) 102, arXiv:2603.18376 [astro-ph.CO]

2026 G. Cerini, **S. Saha** et al., *Lensing-Reconstructed Dark Matter–Intracluster Medium Coherence as a Probe of Cluster Dynamical State: Application to HSTFF, RELICS, and CLASH Clusters*, arXiv:2607.01389 [astro-ph.CO], submitted to ApJ.

2026 L. Ohashi, **S. Saha**, and J. McCleary, *Optimizing Color Selection for Cluster-Member Identification with SuperBIT*, arXiv:2610.06635 [astro-ph.CO], submitted to RNAAS.

2026 A. Field, **S. Saha**, and J. McCleary, *ShearNet: Neural-Network-Based Shear Estimation for Non-Gaussian PSFs*, in preparation.

2024 **S. Saha**, L. Legrand, and J. Carron, *Cluster profiles from beyond-the-QE CMB lensing mass maps*, JCAP 01 (2024) 024, arXiv:2307.11711.

2021 **S. Saha**, S. Shaikh, S. Mukherjee, T. Souradeep, and B. D. Wandelt, *Bayesian estimation of our local motion from the Planck-2018 CMB temperature map*, JCAP 10 (2021) 072, arXiv:2106.07666.

Contributing-Author Publications

2026 G. N. Vassilakis, J. McCleary, M. Amit, **S. Saha**, *Shear Mapping in Python (SMPy): Modular, Extensible, and Accessible Dark Matter Mapping*, *J. Open Source Softw.* 11 (2026) 10928, arXiv:2611.00225 [astro-ph.IM].

2026 P. Voyer, M. Amit, S. J. Benton et al. (including **S. Saha**), *Multi-sensor fusion for fine-guidance and milliarcsecond-level attitude estimation of balloon-borne telescope*, Proc. SPIE 14152, 1415229, arXiv:2607.28816 [astro-ph.IM].

2026 E. Paracha, S. Benton, A. A. Fraisse et al. (including **S. Saha**), *Starlink takes flight: high-bandwidth communication for scientific ballooning*, Proc. SPIE 14155, 141551T.

2026 E. Paracha, S. Benton, A. A. Fraisse et al. (including **S. Saha**), *From sky to screen: improving file downlinking over error-prone, omni-directional links from the stratosphere*, Proc. SPIE 14147, 141473Z.

2026 C. Trendafilova, S. Raghunathan, B. Wallisch et al. (including **S. Saha**), *Sensitivity of Next-Generation CMB Surveys to Neutrinos and Other Light Relics*, arXiv:2608.07453 [astro-ph.CO], submitted to JCAP.

Experiments & Collaborations

SuperBIT Lead developer of the science pipelines, leading multiple key projects.

LSST DESC Involved in multiple projects, focused on cluster weak-lensing systematics

CMB-S4 Member of the Cluster Working Group; contributing to forecasts of galaxy cluster mass estimation.

CMB-Bharat Web Coordinator for the proposed Indian CMB space mission.

Awards & Fellowships

2022 – 2023 Swiss Government Excellence Scholarship (Research), ESKAS No. 2022.0316, hosted by the Université de Genève (CHF 23,540 stipend + CHF 600 travel allowance).

2023 Grant from the Société académique de Genève (SACAD, 2000 CHF) for conference travel.

2023 Infosys Foundation Travel Award and Fellowship (50,000 INR) for conference travel.

2019 Institute PhD Fellowship, IISER Pune (MHRD).

2017 – 2019 Institute M.Sc. Fellowship, IISER Pune (MHRD).

2017 State Topper, National Graduate Physics Examination (top 1% of 774 candidates), Indian Association of Physics Teachers.

2014 – 2017 DST-INSPIRE Scholarship, Ministry of Human Resource Development, Government of India.

Talks & Presentations

Invited Talks

Oct 2026 Seminar. Cosmology Journal Club, Duke University, October 15, 2026

Oct 2026 Seminar. Galaxy Cluster Seminar, University of Michigan, October 06, 2026

Sep 2026 Seminar. BNL Seminar, Brookhaven National Laboratory, Sep 24.

Apr 2026 Seminar. CCAPP Seminar, Ohio State University, Apr 21.

Apr 2026 Seminar. CMB Lunch Talk, Lawrence Berkeley National Laboratory, Apr 17.

Apr 2026 Seminar. KIPAC Tea Talk, Stanford University, Apr 14.

Apr 2026 Seminar. Dark-Sector Meeting, Jet Propulsion Laboratory, NASA, Apr 9.

Apr 2026 Seminar. Cosmology Journal Club, Caltech, Apr 7.

Contributed Talks & Posters

July 2026 Firetalk & poster. LSST DESC Collaboration Meeting, Boston University.

July 2026 Firetalk & poster. Roman workshop, Caltech/IPAC.

Jun 2025 Firetalk & poster. mm Universe 2025, University of Chicago.

Jul 2024 Talk. Relativistic Effects and Novel Observables in Cosmology, Université de Genève.

Nov 2023 Talk. Neighbourhood Cosmology Meeting, Raman Research Institute, Bengaluru.

May 2023 Talk. Third EuCAPT Annual Symposium, CERN, Geneva.

Apr 2023 Talk. Future Cosmology, IESC Cargèse.

Apr 2023 Talk & poster. Future Science with CMB × LSS, YITP Kyoto.

Jan 2023 Talk. Cosmoglobe Winter Workshop, University of Oslo.

Schools & Workshops

Jan 2022 Physics of the Early Universe, International Centre for Theoretical Sciences (ICTS), Bengaluru.

Aug 2021 Data Analysis in Cosmology and Astroparticle Physics, Indian Statistical Institute, Kolkata.

Jun 2021 Summer School in Statistics for Astronomers, Penn State University.

Mar 2019 Pune–Mumbai Collider Meet, IISER Pune.

Computational Skills

Languages	Python, Fortran 90, C/C++, Shell scripting.
Astronomy	GalSim, ngmix, SWarp, SExtractor, PSFEx, PIFF, astropy, plancklens, CAMB, HEALPix/healpy.
ML tools	PyTorch, JAX, TensorFlow, scikit-learn.
Statistical Methods	MCMC sampling (Metropolis–Hastings, Hamiltonian Monte Carlo)
Infrastructure	Git, HPC/SLURM, Linux, LaTeX.
Open-Source Software Contributions	`superbit-lensing` – Primary science pipeline for processing SuperBIT telescope images. `SMPy` – Contributed to mass map reconstruction tool from shear measurements. `ShearNet` – Neural-network-based galaxy shape measurement. `CosmoHMC` – a python Hamiltonian Monte Carlo sampler for field-level inference. `LensIt` – developed the `clusterlens` module.

Mentoring

Research Mentoring

Aug 2025 – Present	**Leila Ohashi**: led to a first-author publication (arXiv:2610.06635 [astro-ph.CO]) and a significant contribution to arXiv:2610.07244 [astro-ph.CO], earning co-authorship.
May, 2025 – Present	**Adam Field** (undergraduate, Worcester Polytechnic Institute): neural-network shear estimation (ShearNet), leading to a first-author publication nearing submission.
2024 – 2025	**Maya Amit** (Northeastern Co-op program): significant contribution to arXiv:2610.07244 [astro-ph.CO], leading to second authorship, and co-authorship on arXiv:2603.18376 [astro-ph.CO].

NSF Young Scholars Program, Northeastern University

Jun – Jul 2026	Mentor to Karuna Tarafdar and Adam Aoua. Completed project; presented poster and talk.
Jun – Jul 2025	Mentor to Leila Ohashi and Marcus Michaud. Completed project; presented poster and talk.

Teaching

Fall 2019	Teaching Assistant, *Electricity & Magnetism* (PHY201), IISER Pune.
Spring 2020	Teaching Assistant, *Nuclear & Particle Physics* (PHY422), IISER Pune.
Fall 2020	Teaching Assistant, *Group Theory in Physics* (PHY356), IISER Pune.

Outreach

IISF 2023	Represented Raman Research Institute (RRI) at the India International Science Festival; team received the Best Conceptual Pavilion Award.
NCSC 2023	Served as a judge for the National Children's Science Congress 2023 (Regional level) in Bengaluru.
RRI Schools	Delivered a talk as a part of the outreach program for school children organised by Raman Research Institute (RRI). Talk - **The Universe in your pocket** [slides].

Service

CMB-Bharat	One of the lead organizers of the Monthly Cosmology Seminar Series for CMB-Bharat, an Indian cosmology consortium (YouTube).
Journal Clubs	Organised the Astrophysics, Cosmology & Particle Physics Journal Club at IISER Pune (2019–2020). Regular participant in the Cosmology Journal Club (Université de Genève) and Very Sirius Meeting (RRI).

Language Proficiency

English (fluent), Bengali (native), Hindi (fluent).